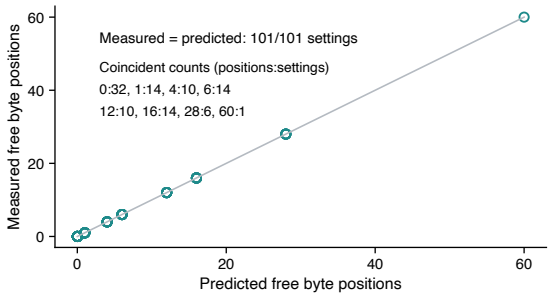
a Registered: 101 settings, 22 parents



Overlapping settings: coordinate counts above; 22 controls /101
Free = arbitrary byte alternatives jointly inherited with causal roots
Fixed core; 22 parents; diagonal = equality

b Direct challenges: all passage directions

Edit / challenge	Cases	Passed
forward / erased	346,290	346,290
forward / joint	6,464	6,464
forward / single	181,050	181,050
reciprocal / erased	486,540	486,540
reciprocal / joint	6,464	6,464
reciprocal / single	40,800	40,800
Total	1,067,608	1,067,608

Reciprocal = restore original displacement; single = 1 site
Joint = payload jointly varied; erased = lost-coordinate control
Pass: exact offspring, ancestry and fixed core; 17 partners/case

c Actual descendants: lineage 0, generation 8

Parent byte 0 = 0x04; copy offset 4 bytes

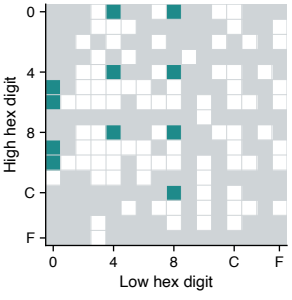
```
04 5E ED B0 04 5E ED B0 04 5E ED B0 04 5E ED B0
04 5E ED B0 04 5E ED B0 04 5E ED B0 04 5E ED B0
```

Parent byte 0 = 0x60; copy offset 32 bytes

```
60 5E ED B0 04 5E ED B0 04 5E ED B0 04 5E ED B0
04 5E ED B0 04 5E ED B0 04 5E ED B0 04 5E ED B0
```

Byte positions 0–31; hex rows = actual descendant bytes
Grey core | red original-parent ancestry lost | teal retained

d Exploratory: all 256 first-byte values



157 nonworking: generation-2 normalized culture gain <0.3
88 working (gain ≥0.3), excluded by prefix/offset/phase
11 working (gain ≥0.3), meets prefix/offset/phase conditions

e Exploratory: sampled-score exceptions

Measurement	Cells
Motif-bearing cells	2347
Cells with dividing offset	2343
Cells with extra fetched-site scores	53
Cells failing whole-tape match gate	1

53 and 1 are disjoint cell subsets of the 2,343 cases

Extra = fetched-site retention beyond unexecuted sites

Gate deficit: 4 alleles retained; offspring <75% parent match

Exploratory Z80 G/I/K/L cells: 5E ED B0 motif (same family as c)
2,347 sampled cells nested in worlds; 4 nondividing excluded

f Frozen candidate eligibility

Eligibility / reason	Candidates
nondividing or out of range	4
Eligible	2343

Same motif-cell candidates as e: d divides L, 4≤d≤L

d = copy offset; L = tape length in bytes

Serial mismatch count 0 /25,856 descendant-transfer records

Each record checked separately for exact bytes, core and ancestry
Shared denominator for each check; conditional overlap family